

## SAXNAS V, Sweden

WMO: 022330

Lat: 64.9678N

Lon: 15.3791E

Elev: 545

StdP: 94.95

Time Zone: 1.00 (EUC)

Period: 01-16

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -28.1      | -24.6 | -30.2                         | 0.2 | -28.1 | -26.8 | 0.3 | -24.5 | 8.2                   | -7.9 | 7.3 | -7.2 | 0.8                   | 230  | 0.698 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 8.7                    | 23.5            | 14.3 | 21.3 | 13.5 | 19.4 | 12.7 | 16.1                | 19.9 | 14.8 | 18.5 | 13.7 | 17.5 | 2.3                  | 270  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 14.5                            | 11.0 | 17.8 | 13.4 | 10.2 | 16.0 | 12.2 | 9.5 | 14.8 | 47.0          | 20.4 | 43.2 | 18.7 | 40.3 | 17.5 | 20.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (d)                        | (e)   | (f)                | (g) | (h)                                                | (i)   | (j)        | (k)   | (l)        | (m)   | (n)        | (o)   | (p)  | (q) |
| 6.9               | 6.0  | 5.2 | DB                         | -32.9 | 25.7               | 2.6 | 1.9                                                | -34.8 | 27.1       | -36.3 | 28.2       | -37.8 | 29.3       | -39.6 | 30.6 |     |
|                   |      |     | WB                         | -33.0 | 16.4               | 2.4 | 1.2                                                | -34.8 | 17.3       | -36.2 | 18.0       | -37.6 | 18.7       | -39.4 | 19.5 |     |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 1.1    | -9.8  | -9.2  | -5.7  | -0.2  | 4.7   | 10.2  | 13.5  | 12.3  | 7.0   | 1.2   | -3.9  | -7.6 |
|                                                                                  | DBStd       | 9.40   | 7.04  | 6.90  | 5.28  | 3.55  | 3.28  | 3.53  | 2.98  | 2.87  | 3.17  | 3.78  | 5.16  | 7.01 |
|                                                                                  | HDD10.0     | 3499   | 613   | 537   | 488   | 307   | 171   | 39    | 4     | 9     | 97    | 273   | 417   | 545  |
|                                                                                  | HDD18.3     | 6293   | 872   | 771   | 746   | 557   | 424   | 243   | 152   | 188   | 339   | 531   | 667   | 803  |
|                                                                                  | CDD10.0     | 252    | 0     | 0     | 0     | 0     | 5     | 46    | 113   | 79    | 8     | 0     | 0     | 0    |
|                                                                                  | CDD18.3     | 3      | 0     | 0     | 0     | 0     | 0     | 0     | 3     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 51     | 0     | 0     | 0     | 0     | 2     | 12    | 32    | 6     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH26.7     | 1      | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 2.3    | 2.2   | 2.3   | 2.4   | 2.4   | 2.4   | 2.5   | 2.3   | 2.2   | 2.4   | 2.6   | 2.2   | 2.2  |
| Precipitation                                                                    | PrecAvg     | 770    | 69    | 52    | 45    | 37    | 42    | 71    | 102   | 90    | 70    | 66    | 62    | 62   |
|                                                                                  | PrecMax     | 1038   | 126   | 126   | 81    | 86    | 86    | 126   | 209   | 191   | 190   | 123   | 141   | 138  |
|                                                                                  | PrecMin     | 553    | 13    | 7     | 7     | 9     | 6     | 12    | 21    | 33    | 28    | 27    | 23    | 25   |
|                                                                                  | PrecStd     | 130    | 27    | 28    | 19    | 16    | 23    | 27    | 48    | 47    | 37    | 26    | 32    | 25   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 3.9   | 3.7   | 6.5   | 11.7  | 20.6  | 24.8  | 26.4  | 24.3  | 17.8  | 10.1  | 7.6   | 4.0  |
|                                                                                  |             | MCWB   | 2.3   | 2.2   | 2.5   | 6.0   | 12.0  | 12.9  | 17.5  | 14.7  | 11.9  | 7.3   | 5.6   | 2.4  |
|                                                                                  | 2%          | DB     | 2.6   | 2.6   | 3.9   | 9.0   | 16.6  | 22.3  | 23.9  | 22.3  | 14.7  | 8.6   | 5.1   | 3.0  |
|                                                                                  |             | MCWB   | 1.1   | 1.3   | 1.6   | 4.0   | 9.0   | 12.1  | 15.8  | 14.3  | 10.6  | 6.7   | 3.5   | 1.6  |
|                                                                                  | 5%          | DB     | 1.2   | 1.3   | 2.9   | 6.7   | 13.5  | 19.6  | 21.6  | 20.3  | 13.0  | 7.6   | 3.6   | 2.0  |
|                                                                                  |             | MCWB   | 0.0   | 0.0   | 1.1   | 2.8   | 7.2   | 11.7  | 14.1  | 13.7  | 9.9   | 6.0   | 2.4   | 0.7  |
|                                                                                  | 10%         | DB     | -0.7  | -0.1  | 1.8   | 5.1   | 10.5  | 16.9  | 19.6  | 18.2  | 11.7  | 6.4   | 2.1   | 0.9  |
|                                                                                  |             | MCWB   | -1.6  | -1.4  | 0.3   | 2.0   | 5.9   | 10.8  | 13.0  | 12.8  | 9.2   | 5.1   | 1.1   | -0.1 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 2.3   | 2.4   | 3.0   | 6.2   | 13.2  | 14.6  | 18.7  | 17.2  | 13.4  | 8.4   | 6.0   | 2.6  |
|                                                                                  |             | MCDB   | 3.5   | 3.5   | 5.3   | 10.9  | 19.9  | 20.2  | 25.3  | 20.2  | 15.6  | 9.5   | 7.1   | 3.6  |
|                                                                                  | 2%          | WB     | 1.2   | 1.5   | 2.0   | 4.5   | 10.0  | 13.5  | 17.0  | 15.4  | 11.7  | 7.1   | 3.9   | 1.7  |
|                                                                                  |             | MCDB   | 2.3   | 2.5   | 3.6   | 7.6   | 14.4  | 18.8  | 22.1  | 19.5  | 13.6  | 8.0   | 4.6   | 2.8  |
|                                                                                  | 5%          | WB     | 0.1   | 0.1   | 1.2   | 3.5   | 7.9   | 12.4  | 15.6  | 14.4  | 10.7  | 6.3   | 2.5   | 0.9  |
|                                                                                  |             | MCDB   | 1.3   | 1.1   | 2.7   | 6.0   | 12.1  | 17.8  | 18.9  | 18.4  | 12.3  | 7.3   | 3.4   | 1.9  |
|                                                                                  | 10%         | WB     | -1.5  | -1.2  | 0.3   | 2.6   | 6.5   | 11.4  | 14.3  | 13.5  | 9.8   | 5.2   | 1.1   | 0.0  |
|                                                                                  |             | MCDB   | -0.8  | -0.2  | 1.7   | 4.7   | 9.9   | 16.3  | 17.4  | 17.1  | 11.2  | 6.4   | 1.9   | 0.8  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 7.3   | 8.4   | 9.6   | 8.4   | 8.4   | 8.7   | 8.7   | 8.5   | 6.3   | 4.8   | 5.1   | 6.3  |
|                                                                                  |             | MCDBR  | 5.2   | 6.7   | 8.7   | 10.8  | 14.4  | 13.5  | 13.0  | 12.6  | 9.1   | 5.7   | 4.9   | 4.9  |
|                                                                                  |             | MCWBR  | 4.8   | 5.9   | 6.7   | 6.3   | 7.4   | 5.4   | 5.5   | 5.2   | 5.2   | 4.3   | 4.2   | 4.5  |
|                                                                                  | 5% WB       | MCDBR  | 5.7   | 6.5   | 7.9   | 8.9   | 12.4  | 11.5  | 10.8  | 11.1  | 7.6   | 5.4   | 4.6   | 4.8  |
|                                                                                  |             | MCWBR  | 5.3   | 6.0   | 6.3   | 5.5   | 7.2   | 5.1   | 5.5   | 5.5   | 5.2   | 4.4   | 4.2   | 4.5  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.175  | 0.249 | 0.280 | 0.310 | 0.319 | 0.319 | 0.330 | 0.318 | 0.293 | 0.272 | 0.345 | 0.179 |      |
|                                                                                  | taud        | 1.935  | 2.190 | 2.243 | 2.276 | 2.444 | 2.521 | 2.504 | 2.523 | 2.578 | 2.429 | 2.811 | 1.951 |      |
|                                                                                  | Ebn at Noon | 362    | 660   | 804   | 862   | 891   | 899   | 875   | 855   | 808   | 645   | 348   | 106   |      |
|                                                                                  | Edh at Noon | 22     | 54    | 82    | 101   | 95    | 90    | 89    | 80    | 61    | 43    | 21    | 7     |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.11   | 0.57  | 1.94  | 3.77  | 4.71  | 4.98  | 4.64  | 3.47  | 1.90  | 0.73  | 0.18  | 0.04  |      |
|                                                                                  | RadStd      | 0.01   | 0.12  | 0.23  | 0.31  | 0.36  | 0.42  | 0.56  | 0.36  | 0.24  | 0.09  | 0.03  | 0.00  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (47) | Regional (37 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page